Highlights

- Low head error (<6%) masks Darcy-flux failure in 75–91% of test runs.
- Wavenumber-squared (k2) weighting amplifies high-frequency head errors.
- Medium roughness invariants screen regime reliability before training.
- Conservative face-flux loss cuts FNO velocity error by 52–95% across media.
- One-cycle AMG defect correction cuts flux error 93% and restores 94% arrival.